

Benchmark Report — Fast-SCNN Floor Segmentation

1. Hardware Tested

The Fast-SCNN floor segmentation model was evaluated on an ASUS V16 laptop with the following hardware:

- **CPU:** Intel® Core™ 7 240H @ 2.50 GHz
- **RAM:** 16 GB
- **GPU:** NVIDIA GeForce RTX 5060 Laptop GPU, 8 GB VRAM
- **Intel GPU:** Available and detected by OpenVINO
- **NPU:** Not available on the tested system

OpenVINO detected the following devices:

CPU, GPU.0, GPU.1

The Intel GPU backend was tested, but the Fast-SCNN OpenVINO graph could not be compiled because of an unsupported GPU layout operation. Therefore, the final OpenVINO benchmark was performed on the **Intel CPU**.

No Intel NPU was available on the tested system, so NPU inference was not benchmarked. The deployment implementation includes device-selection logic to use an NPU when available, with CPU fallback.

2. Segmentation Quality

A custom indoor floor segmentation dataset was used for training and evaluation. Only one foreground class was manually annotated:

Floor

The remaining regions represent non-floor/background areas.

The dataset was divided into training, validation, and test splits. The model was trained using **512 × 512** input images for 100 epochs.

On the held-out validation set, the model achieved:

- **Pixel Accuracy:** 94.9%
- **mIoU:** 87.8%

These results demonstrate that the trained Fast-SCNN model can effectively identify floor regions in the evaluated indoor scenes.

3. OpenVINO FP32 Benchmark

The final OpenVINO deployment was evaluated on the Intel CPU using 512×512 input frames. The end-to-end video pipeline consists of preprocessing, model inference, and postprocessing.

Processing Stage	Latency
Preprocessing	8.86 ms
Inference	18.17 ms
Postprocessing	11.41 ms
Total Latency	38.44 ms/frame

The measured throughput was:

Average Latency: 38.44 ms/frame

Average Throughput: 26.02 FPS

Equivalent Rate: 26.02 Hz

The required deployment target is **at least 2 Hz**, corresponding to a maximum latency of 500 ms per frame. The OpenVINO CPU implementation therefore comfortably exceeds the required real-time performance target.

4. Runtime Comparison

Additional benchmarks were performed using PyTorch, ONNX Runtime, and OpenVINO.

Runtime	Device	Total Latency	Throughput
PyTorch	NVIDIA RTX 5060	21.76 ms/frame	45.96 FPS
PyTorch	Intel CPU	120.11 ms/frame	8.33 FPS
ONNX Runtime	CPU	9.38 ms*	106.63 FPS*
OpenVINO FP32	Intel CPU	38.44 ms/frame	26.02 FPS
OpenVINO INT8	Intel CPU	41.41 ms/frame	24.15 FPS

* The ONNX Runtime benchmark measures the model inference call only and does not include the complete preprocessing and postprocessing pipeline. Therefore, the ONNX

Runtime result should not be directly compared with the end-to-end OpenVINO video benchmark.

5. PyTorch Benchmark

The native PyTorch implementation was evaluated on the NVIDIA RTX 5060 Laptop GPU.

Preprocessing: 8.95 ms

Inference: 7.28 ms

Postprocessing: 5.54 ms

Total Latency: 21.76 ms/frame

Throughput: 45.96 FPS

For comparison, running the same PyTorch pipeline on the Intel CPU resulted in:

Preprocessing: 16.70 ms

Inference: 94.77 ms

Postprocessing: 8.65 ms

Total Latency: 120.11 ms/frame

Throughput: 8.33 FPS

This demonstrates the significant performance improvement obtained from GPU acceleration for the native PyTorch implementation.

6. ONNX Runtime Benchmark

The trained PyTorch model was exported to ONNX and evaluated using ONNX Runtime with the CPU execution provider.

The measured result was:

Inference Latency: 9.38 ms

Inference Throughput: 106.63 FPS

Execution Provider: CPUExecutionProvider

This measurement represents model inference latency only and does not include the complete video processing pipeline.

7. INT8 Quantization

Post-training INT8 quantization was evaluated using NNCF.

The OpenVINO model inference latency changed from:

FP32: 18.17 ms

INT8: 14.65 ms

This represents approximately a 19% reduction in model inference latency.

However, the complete video pipeline became slightly slower:

FP32: 38.44 ms/frame

INT8: 41.41 ms/frame

The increase was mainly due to preprocessing and postprocessing overhead remaining significant portions of the overall pipeline.

Therefore, **OpenVINO FP32 was selected as the final deployment configuration.**

8. Final Results

The Fast-SCNN model achieved the following segmentation quality on the held-out validation set:

- **Pixel Accuracy:** 94.9%
- **mIoU:** 87.8%

The final OpenVINO CPU deployment achieved:

- **Total Latency:** 38.44 ms/frame
- **Throughput:** 26.02 FPS
- **Equivalent Rate:** 26.02 Hz

This substantially exceeds the required deployment target of **2 Hz / 500 ms per frame.**

The results demonstrate that Fast-SCNN provides an effective lightweight solution for floor segmentation and can be successfully converted and deployed using OpenVINO on Intel CPU hardware.